

EXPERIMENT 3

Case Study: Test Cases and Maintenance Plan for an EV Battery Health and Transparency System

Aim

To design test cases and prepare a maintenance plan for an EV Battery Health and Transparency System to ensure that battery information, range predictions, user data, alerts, and system functions operate correctly and reliably.

Objectives

- Identify important functions that require testing.
- Design test cases for the EV battery monitoring application.
- Verify battery health and range calculations.
- Test charging and battery data processing.
- Identify possible software errors and failures.
- Prepare a maintenance plan for long-term system reliability.
- Ensure accurate and user-friendly battery information.

System Under Testing

The system provides EV users with Battery Health (SOH), Battery Percentage (SOC), Estimated Driving Range, Battery Degradation, Charging History, Charge Cycle Count, Battery Temperature, Battery Health Alerts, and Charging Recommendations.

Testing Types

Testing Type	Purpose
Unit Testing	Tests individual software modules
Integration Testing	Tests communication between modules
System Testing	Tests the complete application
Functional Testing	Verifies system functions
Performance Testing	Checks response and loading time
Security Testing	Protects user and battery data
Usability Testing	Ensures the application is easy to use
User Acceptance Testing	Verifies that the system meets user requirements

Test Case Design

Test ID	Test Scenario	Input	Expected Result
TC01	User Login	Valid username/password	Login successful
TC02	Invalid Login	Incorrect password	Error message displayed
TC03	Battery Percentage	SOC = 75%	Dashboard displays 75%
TC04	Battery Health	SOH = 90%	Battery health displays 90%
TC05	Range Calculation	SOC and SOH data	Correct estimated range displayed

Test ID	Test Scenario	Input	Expected Result
TC06	Charging Record	New charging session	Record saved successfully
TC07	Temperature Alert	Temperature above threshold	Warning generated
TC08	Degradation Update	New battery data	Degradation graph updated
TC09	Maintenance Alert	Battery health below threshold	Maintenance notification displayed
TC10	Logout	User selects logout	User successfully logged out

Battery Degradation Test Cases

Test ID	Condition	Expected Result
BD01	New battery	SOH approximately 100%
BD02	Moderate cycle count	Small capacity reduction
BD03	High cycle count	Noticeable degradation
BD04	High temperature	Degradation warning
BD05	Repeated fast charging	Charging impact displayed
BD06	Low battery health	Maintenance recommendation

Range Prediction Testing

The system should estimate range using battery capacity, current SOC, battery health, recent energy consumption, and driving conditions.

Example: Battery Capacity = 60 kWh; Current SOC = 80%; Estimated efficiency = 6 km/kWh. Available Energy = $60 \times 0.80 = 48$ kWh. Estimated Range = $48 \times 6 = \mathbf{288 \text{ km}}$. The system should display the calculated range appropriately.

Alert Testing

Condition	Expected Alert
Battery temperature too high	High-temperature warning
SOH below defined threshold	Battery health warning
Very low SOC	Low-battery warning
Excessive fast charging	Charging recommendation
Battery degradation increasing rapidly	Maintenance alert

Performance Testing

The application should be tested for dashboard loading speed, API response time, database response, large amounts of battery data, multiple users accessing the system, and graph-rendering performance.

Security Testing

The application should protect user login credentials, vehicle information, battery information, charging history, and personal information. Testing should include authentication, authorization, input validation, secure API communication, and protection against common web vulnerabilities.

Maintenance Plan

Software maintenance is required after deployment to ensure that the application remains accurate, secure, and reliable.

1. Corrective Maintenance

Fix errors discovered after deployment, such as incorrect range calculation, incorrect battery health display, login errors, and graph display problems.

2. Preventive Maintenance

Perform database optimization, code cleanup, security updates, backup management, and performance monitoring to prevent future problems.

3. Adaptive Maintenance

Modify the application when the environment changes, including new EV battery technologies, operating systems, APIs, vehicle data formats, or cloud infrastructure.

4. Perfective Maintenance

Improve existing features based on user feedback, including dashboard design, range prediction, battery analytics, graphs, and notification features.

Maintenance Schedule

Activity	Frequency
Database Backup	Daily
System Monitoring	Daily
Error Checking	Weekly
Security Updates	Monthly / As Required
Performance Optimization	Monthly
Battery Prediction Model Review	Quarterly
Feature Improvements	Based on User Feedback
Complete System Review	Every 6 Months

Maintenance Process

Problem Identified → Problem Analysis → Solution Development → Testing → Deployment → Monitoring → Documentation

Expected Results

Testing should ensure that battery health is displayed correctly, range estimates are calculated correctly, charging records are stored properly, battery alerts are generated under appropriate conditions, user information is protected, the application performs reliably, and maintenance activities keep the system accurate and up to date.

Conclusion

The test cases and maintenance plan ensure that the EV Battery Health and Transparency System remains reliable, accurate, secure, and useful throughout its operational life. Testing verifies major functions including battery health calculation, range prediction, charging analysis, alerts, and user authentication. Corrective, preventive, adaptive, and perfective maintenance ensures that the system can respond to software errors, technological changes, security requirements, and user feedback. Systematic testing and continuous maintenance are therefore essential for providing EV users with trustworthy battery information and reducing uncertainty and range anxiety.